AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM341
Computer Vision

Final Examination

Spring Semester 2025/2026
Date: 9/June/2026

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Number of Exam Pages: **Four (04)** (including this cover sheet)	**Time Allowed: 3 Hours**

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

PART – 1 Total Marks 10

Question 1: Select the correct choice for each of the following statements.

1. What is the primary purpose of a convolutional layer in a CNN?
   A) Feature extraction
   B) Sequence modeling
   C) Memory allocation
   D) Dimension reduction

2. Which algorithm is primarily used for dimensionality reduction?
   A) k-NN
   B) Decision Trees
   C) PCA
   D) SVM

3. Which technique is used to detect corners in an image?
   A) Canny
   B) Sobel
   C) Harris
   D) Laplacian

4. Which of the following filters is best for removing salt and pepper noise?
   A) Gaussian filter
   B) Median filter
   C) Sobel filter
   D) Laplacian filter

5. What does image thresholding do?
   A) Reduces image size
   B) Converts an image into binary based on a value
   C) Enhances contrast
   D) Blurs the image

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PART – 2 Total Marks 15

Question 2: (5 marks)

a) Name two optical flow methods used for motion estimation.
b) Give two applications of optical flow.

Question 3: (5 marks)

a) Define Convolutional Neural Network (CNN) and list three of its applications.
b) What is the primary difference between a Dense layer and a Convolutional layer in a neural network?

Question 4: (5 marks)

a) What is a disparity map**?**

b) Complete the following Python code to compute the disparity map using OpenCV.

```
import cv2
import numpy as np

left_img = cv2.imread('left.png', cv2.IMREAD_GRAYSCALE)
right_img = cv2.imread('right.png', cv2.IMREAD_GRAYSCALE)
```

1)

Create stereo block matching object

stereo = _____

2) Compute disparity

disparity = _____

3) Normalize disparity for display

disp_norm = _____

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PART – 3 Total Marks 25

Question 5: (6 marks)

Describe the architecture and function of the deep neural network (DNN) defined in the following code snippet. Explain the role of each layer and how the data flows through the model.

```
model = models.Sequential([
    layers.Dense(128, activation='relu', input_shape=(28*28,)),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])
```

Question 6: (6 marks)

a) Describe Give the definition of Support Vector Machines (SVM). What are the types of SVM?

b) Consider the following cancer test check set where X is the input features an y is the label

```
X = data.drop('target', axis=1)
y = data['target']
```

Write a python code to

1) Split data into training and testing sets

2) Apply SVM in a) on Cancer check data set
3) Predict the output by applying the model in part 2 on new input I

Question 7: (6 marks)

a) What is object detection in computer vision?
b) Explain the main idea behind YOLOv5 and why it is considered a fast object detection model.
c) List two outputs produced by an object detection model.

Question 8: (7 marks)

a) List three common applications of Principal Component Analysis (PCA).
b) Write a python code
   1) Create a dataset with 3 images represented as rows containing 5 pixels intensity values as follows:
      I1=[10, 20 , 30, 40, 50]
      I2=[5, 15, 25, 35, 45]
      I3=[1, 2, 3, 4, 5]
   2) Initialize PCA with 2 components
   3) Fit the model
   4) Obtain the principle components and the transformed dataset

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End of questions

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